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UAL Counterfactual Lab

A behavioral business simulator for United Airlines Holdings

Shivam Arora | Computer Science & Economics, Villanova University
sarora01@villanova.edu | Class of 2027

The investment question

Can United protect cash flow when fuel costs rise, aircraft arrive late, and competitors change capacity? The useful answer depends on how the airline responds, how quickly it can act, and which constraints bind.

I propose a forward-running model that turns public operating data and source-linked disclosures into distributions of capacity, revenue, margin and cash flow. An investor can change a driver, inspect the response, and trace the result back to the evidence and assumptions that produced it.

> **RESEARCH HYPOTHESIS**
>
> Delayed, constrained airline responses should explain outcomes better than a model that changes costs but holds behavior fixed. The project will test that hypothesis against matched-data baselines, not assume it is true.

WHAT IF?

Apply a shock and compare plausible response paths.

WHAT BREAKS?

Find combinations that breach a cash-flow or thesis threshold.

WHAT MUST BE TRUE?

Identify operating assumptions consistent with a target or valuation.

Why United

United offers public operating and financial disclosures that can be joined to federal traffic and service data. The first release will model four geographic markets and a consolidated financial bridge, rather than individual employees or every flight. [2], [4]

The differentiator is a testable map of where an investment thesis fails: show how that boundary changes when airline responses are allowed, delayed or constrained. Stage 2 will deliver the simulator, source ledger, scorecard and interface.

Status: design proposal. No simulator accuracy, runtime or investment-return result is claimed.

01 / MODEL DESIGN

A monthly operating model with dated commitments, constrained choices and an explicit financial bridge.

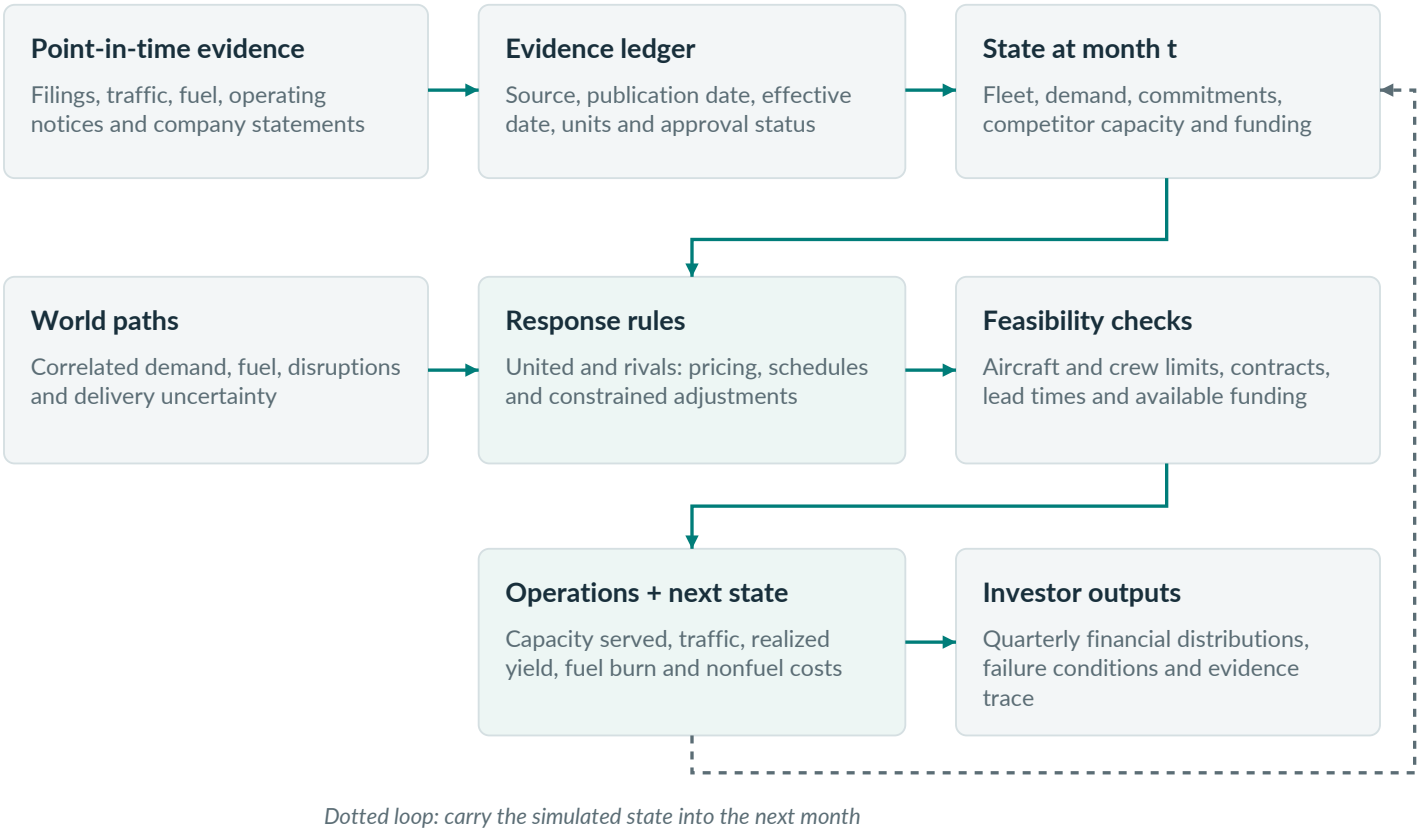


Figure 1. Proposed simulation loop. The LLM structures evidence outside the numerical path loop. Each run preserves its data snapshot, model version and random seed.

Scope	First release
Company boundary	UAL consolidated financials, mainline operations and a separate United Express contract block. Delta, American and other-carrier capacity are lightweight response modules.
Operating resolution	Domestic (U.S. and Canada), Atlantic, Pacific and Latin America. Route-level observations inform market aggregates; no claim of observed route profit.
Time and horizon	Monthly steps; quarterly reporting. Forecast and evaluate 1, 3, 6 and 12 months ahead. Months 13-24 are an explicitly exploratory extension.
Simulation budget	Target 10,000 numerical paths per scenario. Repeat until relevant quantiles stabilize. More paths reduce Monte Carlo error, not model error.

System dynamics provides the stock-and-flow structure. Airline research motivates behavior with adjustment delays, rather than treating United as an unconstrained profit optimizer. [9], [10], [18]

02 / EVIDENCE AND ESTIMATION

Publication time governs what the model may know. Observation frequency is not the same as availability.

Input	Resolution	Role in the simulator
UAL filings and releases [2], [3]	Quarterly results; event-driven updates	Initialize financials, fleet commitments and guidance. Treat management forecasts as claims to test, not realized outcomes.
BTS T-100, Form 41 [4]	Monthly traffic; financial schedules vary	Estimate capacity, traffic, aircraft utilization and costs. Reconcile operating carriers to the UAL reporting boundary.
BTS DB1B / DB1C [5], [6]	Quarterly legacy / monthly newer sample	Estimate fare and traffic mix. DB1C changes sampling in July 2025; test the break rather than concatenate blindly.
Service and weather [4], [8]	Flight-level records; weather observations	Estimate completion and disruption exposure. Use published releases or recorded vintages, not future realized weather.
EIA jet fuel [7]	Daily benchmark, aggregated monthly	Model fuel cost using consumption and a delivered-price basis. Do not substitute crude oil directly for jet fuel.
Unstructured evidence	Dated company and regulator notices	Extract delivery revisions, capacity changes, hub restrictions and contract changes. Use licensed transcripts when available.

AI role

A language model converts documents and natural-language scenarios into a typed record: entity, variable, value or range, units, publication date, effective period, source passage and evidence type. Deterministic checks reject invalid units, dates and contradictions. Uncertain updates require approval. Historical tests admit only verified source extractions, never LLM-authored coefficients or forecasts. Keep the extracted passage for audit.

STATEMENT
Observed / guidance / hypothesis

→

CHECK
Source, units, dates, conflicts

→

MODEL UPDATE
Approved patch and visible diff

Estimated vs. assumed

Fit a parsimonious state-space demand model and lagged response rules, with partial pooling across comparable carrier-market panels. Keep accounting identities fixed. Estimate fare and demand jointly where the data support it; label elasticities as sensitivity assumptions when causal identification is weak. A good historical fit alone does not establish a causal response. [10], [11]

Data gate: core work uses public sources. Paid schedules, booking data and consensus estimates are optional. Flag missing history; never synthesize it as ground truth. Longaeva data require an NDA and approval. [1]

03 / INVESTOR WORKFLOW

The interface exposes assumptions and constraints before it presents a probability.

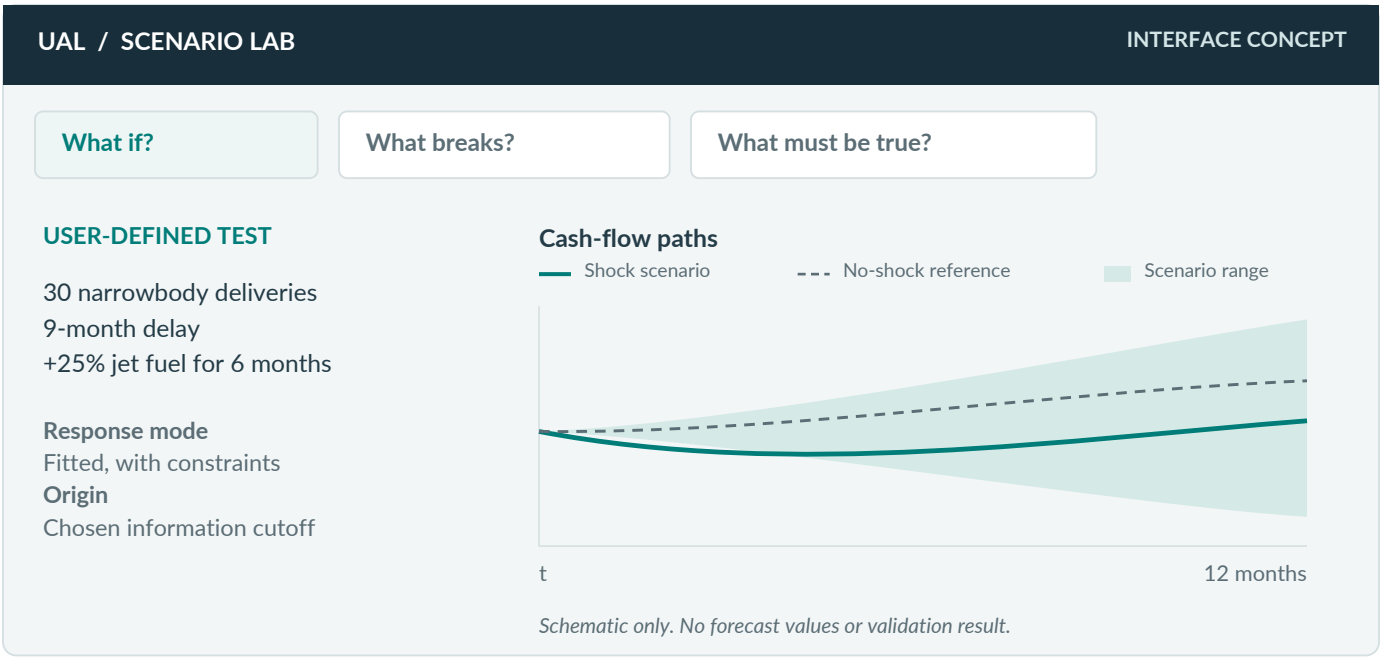


Figure 2. Proposed screen. Selecting an output opens the assumptions, cited evidence, response constraints and source changes behind it.

What if? Delay the delivery queue, not aircraft already in service. Compare the same world paths under frozen behavior and fitted, constrained responses. Track deferred capital spending, retained aircraft costs, schedule changes and cash timing. A delay can reduce growth and preserve near-term cash at the same time.

What breaks? Search joint driver ranges for negative 12-month free cash flow or a user-defined cash floor. Use scenario discovery to summarize failure regions. Report sensitivity to the tested ranges and policy model. A fraction of an exploratory grid is not an event probability. [12]

What must be true? First invert an explicit revenue, margin or cash-flow target. Then add valuation scenarios using disclosed discount-rate and terminal assumptions. Show the set of compatible operating paths, not a unique market forecast. Guidance, analyst consensus and price-implied assumptions remain separate. [16]

THE INVESTOR HANDOFF

Export a one-page thesis record: scenario, distribution, binding constraint, disagreement with the chosen reference, next observable checkpoint and evidence that would overturn the view. No automated trading.

04 / VALIDATION

Forecasts, mechanisms and decision value are tested separately. All thresholds below are acceptance criteria, not results.

Historical validation

Use a release-aware, expanding-window evaluation: 2013-2019 for initial fitting, 2020-2023 for model selection, and 2024 onward for a locked temporal test. Score operations at 1, 3, 6 and 12 months; score financials at 1, 2 and 4 quarters, after outcomes mature. Every cutoff uses information available by that date.

Where historical publication vintages cannot be recovered, mark the exercise as a reconstruction and exclude it from point-in-time performance claims. The temporal test is retrospective. A separate, frozen forward log begins when the prototype is released.

Test	Evidence required
Accounting and feasibility	Reconcile base-period revenue, expenses and cash flow within disclosed rounding tolerances. Zero negative fleet counts, excess served capacity or duplicated regional fuel charges.
Forecast quality	Compare seasonal persistence, a fixed-policy driver model and a regularized statistical forecast using identical information. Report error, CRPS, interval score, coverage and width.
Behavior and interventions	Test response direction, lag and magnitude around held-out disruptions. Use placebos and parameter-recovery tests. Replay with known future shocks is a mechanism test, not a forward forecast.
Incremental contribution	Remove competitor feedback, unstructured updates and regime calibration separately. Retain complexity only when it adds forecast skill or a defensible scenario capability.
Reliability and usefulness	Reproduce runs from a manifest; test stale inputs, extreme shocks and unsupported prompts. Report runtime and evidence-extraction errors. Log analyst task completion and revisions.

Uncertainty decomposition

Draw uncertain parameters once per path, then simulate correlated world shocks over time. Compare plausible response models to expose structural disagreement. Report Monte Carlo error separately. An arbitrary ensemble of model forms is not a calibrated probability distribution.

Use regime-aware conformal calibration only for forecast quantities with enough resolved observations. Publish sample counts, interval widths and block-based uncertainty estimates by horizon and regime. Pool sparse groups or display insufficient evidence. No joint-path coverage or extreme-intervention guarantee is claimed. [13], [14], [15]

Investment test: freeze a paper decision rule before evaluation, include trading and hedging costs, and compare with no-signal and sector benchmarks. Better business forecasts are not, by themselves, proof of alpha.

05 / BUILD PLAN

Stage 2 targets one focal company, a bounded simulator and a reproducible evidence package.

Milestone	Deliverable
Week 3 Foundation	Publish the source manifest and carrier mapping. Reconcile financial and operating baselines. Implement fleet, fuel and capacity mechanics; run one scenario end to end.
Week 4 Behavior and interface	Fit demand and response modules. Add source-checked text updates, paired paths and the scenario screen. Use the mentor check-in to challenge assumptions and data gaps.
Week 5 Prototype return	Deliver all three workflows, a versioned model card, baseline/ablation scorecard, reproducible commands and a hosted or local demo. Include failed tests and unsupported cases.
Weeks 6-7 Finals preparation	Exercise unseen prompts with users, record latency and usability, and refine the failure-region summaries. Prepare the live demonstration, implementation plan and scaling roadmap.

Build and operating cost

Build with Python, DuckDB/Parquet and React. Preserve immutable data snapshots and cache approved LLM extractions outside the path loop. Target a 10-second cached scenario on a standard multicore CPU; report latency percentiles and cost per update. The core design needs no GPU or paid booking feed.

If coverage or runtime fails the gate, simplify response rules and widen assumption ranges. Keep the accounting model, source trace and limitations intact. New airlines should require data mappings and policy recalibration, not a new simulation service.

Ownership and data controls

I will own the implementation, evidence review and model decisions. Third-party sources and code will be recorded with terms and attribution. Restricted data remain isolated from the public demo and external model providers unless explicitly approved; API keys stay server-side. AI-assisted work will be disclosed with prompts and test records where relevant.

Prior research I led on regime-conditional conformal prediction informs the evaluation approach used here. No fitted models, empirical results, or performance guarantees from that work are reused. All uncertainty calibration in the UAL simulator will be estimated and validated independently on airline data. [17]

Challenge fit

Forward simulation and business drivers: pp. 2, 7-8. **Observable and unstructured evidence:** p. 3. **Stochastic outcomes and interpretability:** pp. 4-5. **Differentiation:** pp. 1, 4-5. **Repeatability and scale:** this page. **Working prototype:** the Week 5 return. The present submission is the Stage 1 written proposal. [1]

APPENDIX A / STATE MODEL

State carried month to month across four UAL geographic markets and two mainline fleet groups. Regional flying remains a separate contract block.

Stored state	Unit	Update rule / boundary
Fleet F[f], U[f]	Aircraft counts	Mainline fleet F includes unavailable aircraft U. Deliveries and retirements change F; outages and returns change U. Available aircraft = F - U.
Commitment queues D[f,h], R[f,h]	Aircraft by future month	Delivery and retirement queues for h = 1...24. A delay shifts deliveries, not aircraft already in service.
Scheduled capacity A[r,h], Areg[r,h]	Seat-miles by month	Mainline and contracted regional plans, with three forward months stored. Disclosed assumptions set revision lead times and disruption costs.
Demand Z[r]	Latent index	Demand inferred from traffic, seasonality and signals. Estimate monthly uncertainty from mixed-frequency observations.
Pricing memory Y[r]	USD per passenger-mile	Lagged all-in passenger yield, with measurement uncertainty. Update from pricing response and traffic mix, jointly with demand.
Competition C[k,r]	Seat-miles	Capacity for k = Delta, American and residual competitors. Lagged responses; widen assumptions where international coverage is weak.
Liquidity B	USD	Unrestricted cash and equivalents, rolled forward with cash flows. Liquid investments are tracked separately.
Funding Debt[h], Credit	USD	Debt and finance-lease maturities, months 1-24 plus later debt; committed credit. No unlimited borrowing or assumed refinancing.
Capital assets PPE	USD	Net operating assets for depreciation and capital spending. Keep leased capacity separate from owned-asset accounting.
Operating balances AT, LP, W	USD	Advance ticket sales, loyalty deferrals and other working capital. Cash received is not necessarily revenue earned.

Exogenous inputs and outputs

Jet fuel, macro demand, weather, delivery disruptions and interest rates enter as external paths. Crew and airport limits, wage rates, fuel efficiency, contract costs and nonfuel cost bases are dated parameters or constraint schedules.

Available seat-miles (ASM), revenue passenger-miles (RPM), load factor, revenue, unit costs, margin and free cash flow are derived outcomes, not independent random draws. Every stored input records its source, date, unit and estimation method; missing private information remains a labeled assumption.

APPENDIX B / TRANSITIONS

Each month: read available state, apply the scenario, choose feasible actions, realize demand and operations, then close the financial accounts.

Decision family	Feasibility rule
Capacity and market mix	Adjust future schedules, subject to aircraft, crew, airport access and commitment windows. Emergency cuts carry disruption costs.
Pricing response	Adjust aggregate yield targets with lags and bounds. Demand responds through an explicit elasticity range. Do not infer a booking-level optimizer from aggregate data.
Fleet retention / use	Postpone eligible retirements or change utilization within maintenance and crew limits. New orders beyond existing commitments are outside the first release.
Discretionary cash use	Defer genuinely optional spending and distributions; use committed credit only within disclosed limits. Mandatory maintenance, safety and contracts remain binding.

Transition: $S_{t+1} = T(S_t, a_t, X_{t+1}; \theta)$. Actions a_t follow a fitted policy using only current information, projected into the feasible set. X supplies external shocks; θ holds the path's parameters.

Core identities

Fleet: $F_{t+1} = F_t + \text{deliveries}_t - \text{retirements}_t$
Traffic: $RPM_{r,t} = \min(\text{desired RPM at chosen yield}, \text{served } ASM_{r,t})$
Load factor: $LF_{r,t} = RPM_{r,t} / ASM_{r,t}$; undefined when ASM = 0
Passenger revenue: $R_{pax,t} = \text{sum}_r(RPM_{r,t} \times \text{all-in passenger yield}_{r,t})$
Operating profit: $R_{pax} + R_{cargo} + R_{other}$ - fuel - nonfuel expense
Model free cash flow: FCF = operating cash flow - cash capital expenditures
Liquidity: $B_{t+1} = B_t$ + FCF + net financing + other cash movements

Fuel expense equals gallons consumed times delivered price. The income-to-cash bridge covers advance sales, loyalty deferrals, interest and taxes. Other cash movements include non-capex investing, FX and cash reclassifications. Do not double-count ancillaries or award revenue already in yield, or regional fuel in contract fees. [2]

Scenario contract

Each scenario records target, magnitude, onset, duration, recovery, geography and response mode. Supported families: fuel, demand, deliveries, temporary aircraft/hub availability, rival capacity and contract costs. Preserve specified shock dependence. Unsupported requests return a boundary explanation, not an invented coefficient.

Response modes: frozen policy, fitted observed-like response, and separately labeled feasible-policy sensitivity. Attribute changes through paired runs on common world paths. Report outcome attribution separately from uncertainty decomposition.

Extreme cases: a 10x fuel shock, leadership replacement, layoffs, bankruptcy restructuring or a merger is not a calibrated forecast in v1. Financial stress can breach a cash threshold; that is not a legally modeled bankruptcy event.

SOURCES / 1 OF 2

Primary data and research foundations. Numbered citations link here; source titles link to the original publication. Web sources checked 16 September 2026.

01 **Longaeva. Official AI Hackathon rules.**

Stage 1 written proposal; challenge, evidence, interpretability, scale and competition schedule. Accessed 16 September 2026.

02 **United Airlines Holdings. 2025 Form 10-K.**

Business and regional capacity arrangements; revenue recognition; financial statements and commitments. Fiscal year ended 31 December 2025.

03 **United Airlines. Second-quarter 2026 results.**

Dated management evidence for fuel recovery and capacity responses. Statements about future quarters are guidance, not outcomes. 15 July 2026.

04 **Bureau of Transportation Statistics. Airline Information for Download.**

T-100, Form 41 financial/traffic schedules and on-time reporting. Coverage, definitions and release lags must be checked per series.

05 **Bureau of Transportation Statistics. First Quarter of OD40 Data Is Live.**

DB1C is a monthly 40% ticket sample; the initial July-September 2025 release was announced on 17 February 2026.

06 **Bureau of Transportation Statistics. TranStats Data Index.**

Dataset entry points and definitions for DB1B/DB1C and airline operating data.

07 **U.S. Energy Information Administration. Gulf Coast jet fuel spot price.**

Kerosene-type jet fuel benchmark in dollars per gallon, with daily and aggregated history. A benchmark is not UAL's delivered price.

08 **NOAA National Centers for Environmental Information. Climate Data Online.**

Historical weather observations and station information for disruption covariates.

09 **Pierson, K., and J. D. Sterman (2013). Cyclical dynamics of airline industry earnings.**

System Dynamics Review, 29(3), 129-156. Behavioral capacity, demand and pricing feedbacks; DOI: 10.1002/sdr.1501.

SOURCES / 2 OF 2

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Quarterly Journal of Economics, 139(2), 1149-1199. Evidence against an undifferentiated, unitary pricing decision-maker; DOI: 10.1093/qje/qjad051.

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Journal of Econometrics, 168(1), 156-173. Network decisions, competition and counterfactual assumptions; DOI: 10.1016/j.jeconom.2011.09.013.

12 **Kwakkel, J. H. (2017). The Exploratory Modeling Workbench.**

Environmental Modelling & Software, 96, 239-250. Scenario discovery and robust decision support; DOI: 10.1016/j.envsoft.2017.06.054. Implementation documentation linked.

13 **Gibbs, I., and E. Candès (2021). Adaptive Conformal Inference Under Distribution Shift.**

NeurIPS 34, 1660-1672. Online calibration of coverage frequency, not automatic validation of unseen causal interventions.

14 **Gneiting, T., and A. E. Raftery (2007). Strictly Proper Scoring Rules, Prediction, and Estimation.**

Journal of the American Statistical Association, 102(477), 359-378. Scoring distributions and intervals; DOI: 10.1198/016214506000001437.

15 **Sargent, R. G. (2020). Verification and Validation of Simulation Models: An Advanced Tutorial.**

Winter Simulation Conference, 16-29. Data, conceptual, software and operational validity assessed against intended use.

16 **Mauboussin, M. J., and A. Rappaport (2021). Expectations Investing: Reading Stock Prices for Better Returns.**

Revised edition, Columbia University Press. Reverse valuation as an assumption test, used here without claiming uniquely identified market beliefs.

17 **Arora, S., and V. Margapuri (2026). Regime-Conditional Conformal Prediction for Realized Volatility Forecasting.**

Author-provided research manuscript. Design inputs: time-aware evaluation, regime diagnostics and interval calibration. Airline results require separate validation.

18 **Sterman, J. D. (2000). Business Dynamics: Systems Thinking and Modeling for a Complex World.**

McGraw-Hill. Stocks, flows, feedback and delays. The simulator adapts established modeling principles rather than claiming a new simulation theory.